

Interacting Multipoles of Rank 2 and 4

[illegible]

Simplify deltas:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = R_z^{-13} \cdot \frac{1}{315} \cdot$$

(2a)

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\delta\epsilon\zeta}^A \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\delta\epsilon\epsilon}^A \\
& -1890 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\delta\delta\epsilon}^A \\
& -2835 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\gamma\delta\epsilon}^A \\
& -3780 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& -3780 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\gamma\delta\epsilon}^A \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\gamma\delta\epsilon}^A \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\beta\gamma\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\beta\gamma\gamma\delta}^A \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\gamma\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\gamma\gamma\delta}^A \\
& +630 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\delta}^A \\
& +1260 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\delta}^A \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (2d)
\end{aligned}$$

Simplify R:

$$\begin{aligned}
& + \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zz\epsilon\epsilon}^A \\
& -1890 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zz\delta\delta}^A \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -3780 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Phi_{zzz\beta}^A \\
& -3780 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{zz}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\delta\delta}^A \\
& +945 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\
& +1260 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (3a)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 945 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\epsilon\epsilon}^A \\
& -1890 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\delta\delta}^A - 2835 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\gamma\gamma}^A \\
& -3780 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{zzz\beta}^A - 3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A \\
& -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A + 315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\gamma\gamma\delta\delta}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\delta\delta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\beta\gamma\gamma}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\delta\delta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\gamma\gamma}^A \\
& +630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (3b)
\end{aligned}$$

$$\begin{aligned}
& \xrightarrow{\text{reduced indices}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[\begin{aligned}
& +10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 945 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -1890 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A - 2835 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\
& -3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A - 3780 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A \\
& -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A + 315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A \\
& +315 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A + 945 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A \\
& +630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A \\
& -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A - 180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A
\end{aligned} \right] \quad (3c)
\end{aligned}$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} +10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 5670 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\ -7560 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{zzz\alpha}^A - 945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\ +315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{z\beta}^B \cdot \Phi_{z\alpha\alpha\beta}^A \\ +1260 \cdot R_z^6 \cdot \Theta_{z\alpha}^B \cdot \Phi_{z\alpha\beta\beta}^A + 630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{zz\alpha\beta}^A - 45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ -180 \cdot R_z^6 \cdot \Theta_{\alpha\beta}^B \cdot \Phi_{\alpha\beta\gamma\gamma}^A \end{bmatrix} \quad (3d)$$

Simplify Quadrupoles:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} +10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 5670 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\ -7560 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\ +315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\ +1260 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\beta\beta}^A + 630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A - 45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ -180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\gamma\gamma}^A \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} +10395 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 5670 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\ -7560 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A \\ +315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\ +1260 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A + 630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A \\ +1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A - 45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A \\ -180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \end{bmatrix} \quad (4b)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \begin{bmatrix} +2835 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A - 3150 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zz\alpha\alpha}^A \\ -945 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zzzz}^A + 315 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \\ +630 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\beta\beta}^A + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \\ -45 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\beta\beta\gamma\gamma}^A - 180 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{\alpha\alpha\beta\beta}^A \end{bmatrix} \quad (4c)$$

Exploit Traceless Conditions:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = R_z^{-13} \cdot \frac{1}{315} \cdot \left[+2835 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A + 1260 \cdot R_z^6 \cdot \Theta_{\alpha\alpha}^B \cdot \Phi_{zz\alpha\alpha}^A \right] \quad (5)$$

Multiply Out Sum:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = R_z^{-13} \cdot \frac{1}{315} \cdot \left[+2835 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A + 1260 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A \right. \\ \left. +1260 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A + 1260 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A \right] \quad (6a)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{315} \cdot \left[+4095 \cdot R_z^6 \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A + 1260 \cdot R_z^6 \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A \right. \\ \left. +1260 \cdot R_z^6 \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A \right] \quad (6b)$$

Combining everything:

$$+ \frac{1}{315} \cdot T_{\alpha\beta\gamma\delta\epsilon\zeta} \Theta_{\alpha\beta}^B \Phi_{\gamma\delta\epsilon\zeta}^A = \left[+13 \cdot R_z^{-7} \cdot \Theta_{zz}^B \cdot \Phi_{zzzz}^A + 4 \cdot R_z^{-7} \cdot \Theta_{xx}^B \cdot \Phi_{xxzz}^A \right. \\ \left. +4 \cdot R_z^{-7} \cdot \Theta_{yy}^B \cdot \Phi_{yyzz}^A \right] \quad (7)$$